



Histology of the Pancreas

Histology GIT Block · MEDucation by Mattin Majid

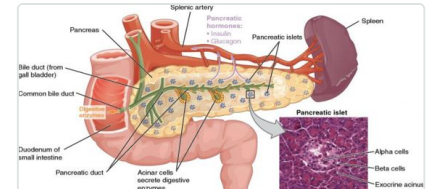
BEFORE YOU READ – 3 BLIND ID CHALLENGES

1. Name the four islet cell types, where each sits in the islet, and what each secretes.
2. Trace a drop of pancreatic secretion from an acinar cell to the duodenum, naming every duct.
3. What is unique about the centroacinar cell, and what stimulates its secretion?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

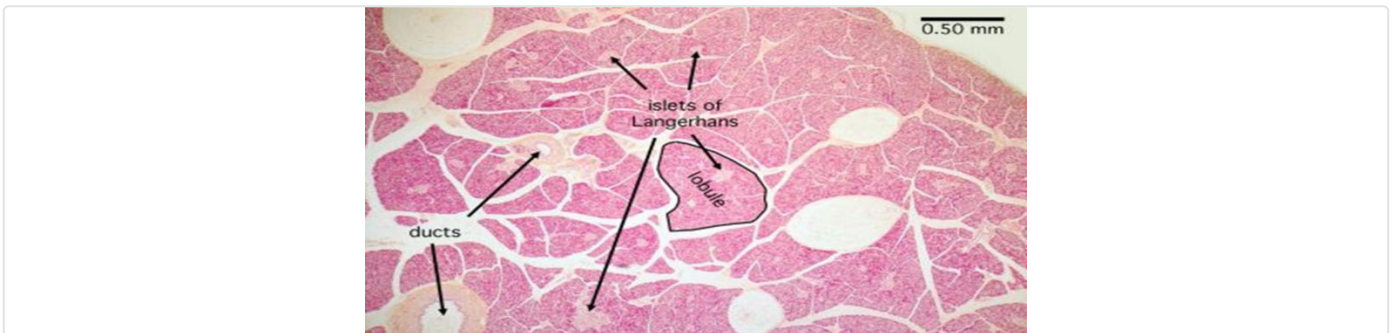
1 · The pancreas — one-line ID rules

- **Shape and position:** elongated, **retroperitoneal**; a large **head** near the duodenum, narrowing to a **body** and **tail** extending left.
- **Mixed gland:** **exocrine** (digestive enzymes, via acini + ducts) and **endocrine** (hormones, from the **islets of Langerhans**).
- A thin **loose CT capsule** surrounds the gland; **septa** extending from it divide the gland into **lobules and lobes** and carry blood vessels and nerves.
- The exocrine part is a **compound acinar gland**; its secretion digests **carbohydrates, proteins and fats**.



Pancreas: head, body, tail, main pancreatic duct, bile duct, duodenum, spleen, acini, pancreatic islet (inset: alpha/beta cells).

Low-power ID: a sea of deeply **eosinophilic, closely packed acini** (exocrine) interrupted by scattered **pale, round islands** of smaller cells (the islets of Langerhans) — unlike the parotid, there is **no serous demilune** and no mucous acini.

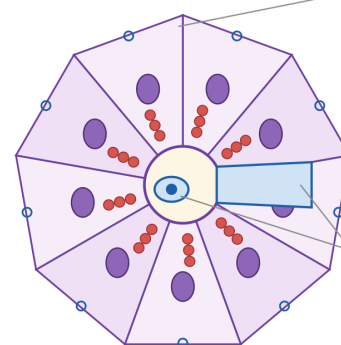


Low power: exocrine acini with pale islets of Langerhans and ducts scattered between them.

2 · Exocrine acinus — the pyramidal cell, step by step

One-line ID: each acinus = several **pyramidal, enzyme-secreting cells** around a tiny lumen; **basal** cytoplasm is **basophilic** (RER + ribosomes), **apical** cytoplasm holds **eosinophilic zymogen granules**.

- Enzymes are packaged and stored in **membrane-bound zymogen granules** in the apical cytoplasm.
- Release of these enzymes is controlled by **cholecystikinin (CCK)**, produced by **enteroendocrine cells of the duodenal mucosa**.



Pyramidal acinar cell
6–8 wedge-shaped cells per acinus

Basal RER
basophilic cytoplasm (protein synthesis)

Nucleus
round, basal third of the cell

Zymogen granules
apical, eosinophilic — stored enzymes

Lumen
centre of the acinus

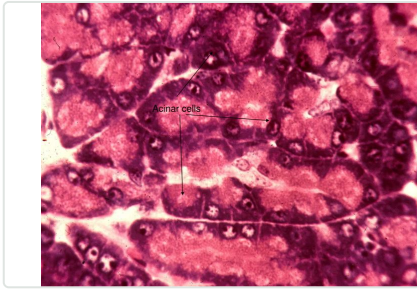
Centroacinar cell
pale, clear cytoplasm, sits in the middle of the acinus

Intercalated duct
begins here — the smallest duct

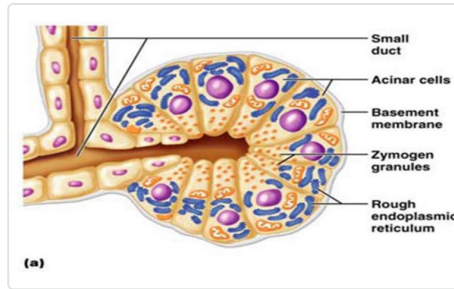
Each acinus has **several** pyramidal cells around one lumen, with 1–2 centroacinar cells.

Anchor diagram: one acinus — basal RER, apical zymogen granules, lumen, centroacinar cell, intercalated duct.

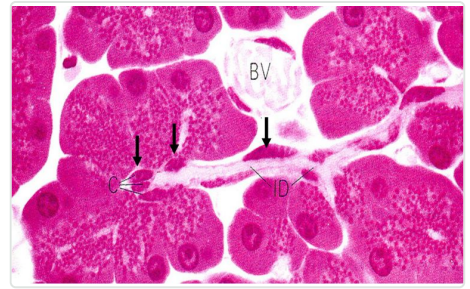
ENZYME GROUP	EXAMPLES	DIGESTS
Amylase	—	carbohydrates
Lipase and related enzymes	phospholipase A2, cholesterol esterase	fats
Proteases	trypsin, chymotrypsin, carboxypeptidase, elastase	proteins



Acinar cells (H&E): basal basophilia, apical eosinophilic granules.



Acinus schematic: small duct, acinar cells, basement membrane, zymogen granules, RER.



Intercalated duct (ID) between acini; blood vessel (BV).

3 • Centroacinar cells and the duct system

One-line ID: centroacinar cells have a **condensed nucleus and clear cytoplasm**; 1-2 sit inside the lumen of each acinus, at the **origin of the intercalated duct** — found **only in the pancreas**, nowhere else in the GIT.

Centroacinar cells secrete a **watery, bicarbonate-rich fluid** in response to **secretin** from enteroendocrine cells of the small intestine. This fluid **neutralizes acidic chyme**, optimizing pancreatic enzyme activity.

THE DUCT PATHWAY — EACH STEP NAMED, LINED AND LOCATED

DUCT	LINING	WHERE / NOTE
1 • Intercalated duct	simple squamous or low cuboidal	smallest duct; begins directly from the centroacinar cells inside the acinus
2 • Intralobular duct	low columnar	formed by several intercalated ducts joining within one lobule ; carries secretion toward larger ducts
3 • Interlobular duct	columnar , may contain goblet cells	found between lobules , in the septa; surrounded by abundant connective tissue
4 • Main pancreatic duct	—	formed by many interlobular ducts converging; runs tail → head ; joins the common bile duct to open at the major duodenal papilla (ampulla of Vater) , regulated by the sphincter of Oddi

Pathway to remember: acinar cells → intercalated ducts → intralobular ducts → interlobular ducts → main pancreatic duct → duodenum.

MAGNIFICATION LADDER (SCAN-PATH ORGANISATION OF THE LECTURE'S FEATURES — THE LECTURE GIVES NO MAGNIFICATIONS)

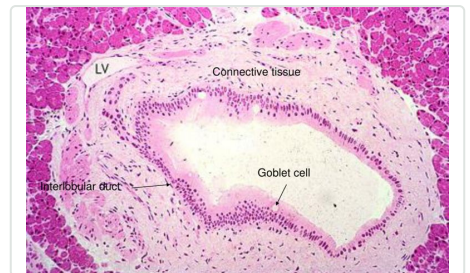
MAG	WHAT TO LOOK FOR
×4	Lobules of eosinophilic acini separated by CT septa carrying interlobular ducts; pale islets scattered between them.
×10	Intralobular ducts (low columnar) within a lobule; interlobular ducts (columnar ± goblet cells) in the septa, with abundant surrounding CT.
×40	Intercalated duct (squamous/low cuboidal) opening from a centroacinar cell at the centre of an acinus; acinar cell basal RER vs apical granules.



Intercalated duct within the acinar tissue.



Intralobular duct: exocrine acinar cells, connective tissue.



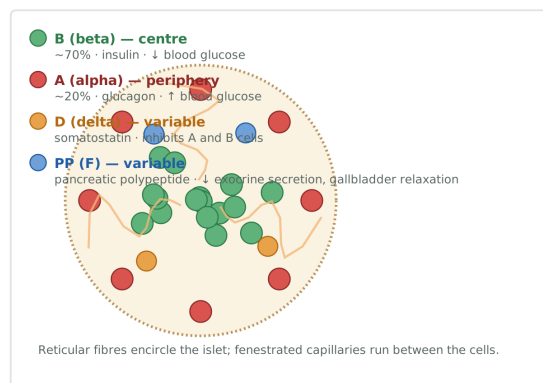
Interlobular duct: connective tissue, goblet cell.

Look-alike pair: centroacinar cell vs duct epithelium. **Centroacinar** = pale, clear, round cell sitting inside the acinar lumen itself (no duct wall yet); **intercalated duct epithelium** = flat-to-cuboidal cells forming the wall once the duct is visibly separate from the acinus.

4 · Endocrine portion — islets of Langerhans

One-line ID: pale, polygonal clusters of cells, **smaller** than the surrounding acinar cells — a round island standing out against the deep pink acinar sea; **most numerous in the tail**.

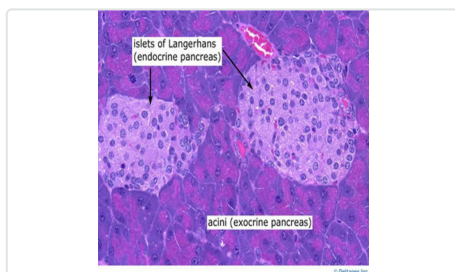
- Cytoplasm holds **secretory granules containing hormones**; cells are separated by a network of **fenestrated capillaries**.
- Each islet is surrounded by **reticular fibres** of connective tissue.



Anchor diagram: one islet of Langerhans — cell types by position and share.

CELL	LOCATION IN ISLET	SECRETES	SHARE	ACTION
A (alpha)	periphery	glucagon	~20%	increases blood glucose
B (beta)	centre	insulin	~70%	decreases blood glucose
D (delta)	variable	somatostatin	—	inhibits glucagon and insulin secretion
PP (F)	variable	pancreatic polypeptide	—	inhibits pancreatic exocrine secretion (enzymes + bicarbonate); relaxes the gallbladder and decreases bile secretion

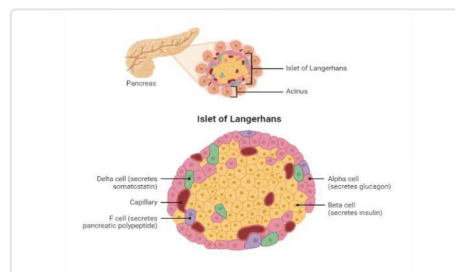
Look-alike pair: exocrine acinus vs islet, side by side on one slide. **Acinus** = deep eosinophilic, cells radiate around a tight lumen, no capillary network visible between cells. **Islet** = pale, polygonal cells in loose clusters threaded by fenestrated capillaries, no lumen.



Islets of Langerhans (endocrine) beside acini (exocrine) — low power contrast.



Islet: secretory acini, connective tissue capsule, capillaries, pancreatic islet cells.



Islet schematic: alpha, beta, delta, F cells and capillary.

Pearls — what must survive if everything else is forgotten

- Acinus:** pyramidal cells, basal basophilia (RER), apical eosinophilic **zymogen granules**; enzyme release is triggered by **CCK** from duodenal enteroendocrine cells.
- Centroacinar cells** are unique to the pancreas: pale, clear, sit inside the acinar lumen at the start of the intercalated duct; **secretin** drives their bicarbonate-rich secretion.
- Duct order:** intercalated (squamous/low cuboidal) → intralobular (low columnar) → interlobular (columnar, ± goblet cells, lots of CT) → main pancreatic duct → ampulla of Vater (sphincter of Oddi).
- Three enzyme groups:** amylase (carbohydrate), lipase + phospholipase A2 + cholesterol esterase (fat), proteases — trypsin, chymotrypsin, carboxypeptidase, elastase (protein).
- Islet cells:** **B (centre, ~70%, insulin, ↓ glucose)**; **A (periphery, ~20%, glucagon, ↑ glucose)**; **D (somatostatin, inhibits A & B)**; **PP/F (↓ exocrine secretion, gallbladder relaxation)**.

Test yourself — transfer-task MCQs

Unseen fields, not slide recall. Commit to an answer before you uncover the key.

1. A biopsy shows pyramidal cells with basophilic cytoplasm near the base and eosinophilic granules near the apex, arranged around a tiny lumen. Which hormone triggers release of the apical granule contents?

- A. Secretin
- B. Cholecystokinin (CCK)
- C. Somatostatin
- D. Gastrin
- E. Insulin

2. A small, pale cell with a condensed nucleus and clear cytoplasm sits within the lumen of a pancreatic acinus, at the point where a duct begins. What does this cell secrete, and in response to what stimulus?

- A. Bicarbonate-rich fluid, in response to secretin
- B. Digestive enzymes, in response to CCK
- C. Insulin, in response to high glucose
- D. Glucagon, in response to low glucose
- E. Mucus, in response to gastrin

3. Tracing pancreatic secretion from an acinus toward the duodenum, which duct comes immediately after the intralobular duct?

- A. Intercalated duct
- B. Interlobular duct
- C. Common bile duct
- D. Ampulla of Vater
- E. Centroacinar duct

4. A pale, polygonal cluster of cells lies among the deeply eosinophilic pancreatic acini, laced with a network of small capillaries. Nearly a third of its cells sit at its periphery. What do those peripheral cells secrete?

- A. Insulin
- B. Glucagon
- C. Somatostatin
- D. Pancreatic polypeptide
- E. Secretin

5. A duct lined by columnar epithelium with scattered goblet cells runs through a connective-tissue septum between two pancreatic lobules. Which duct is this, and what does it connect?

- A. Intercalated duct, connecting acini to centroacinar cells
- B. Intralobular duct, connecting ducts within one lobule
- C. Interlobular duct, connecting intralobular ducts to the main pancreatic duct
- D. Main pancreatic duct, connecting the pancreas to the duodenum
- E. Common bile duct, connecting the gallbladder to the duodenum

ANSWERS – COVER UNTIL YOU'VE COMMITTED

- 1. **B** — CCK, from duodenal enteroendocrine cells, controls release of the zymogen (enzyme) granules from acinar cells.
- 2. **A** — Centroacinar cells secrete a watery, bicarbonate-rich fluid in response to secretin, neutralizing acidic chyme.
- 3. **B** — The order is intercalated → intralobular → interlobular → main pancreatic duct → duodenum.
- 4. **B** — A (alpha) cells, at the periphery of the islet (~20% of cells), secrete glucagon, which raises blood glucose.
- 5. **C** — Interlobular ducts sit in the septa between lobules, are columnar (sometimes with goblet cells) and carry secretions from intralobular ducts toward the main pancreatic duct.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam

Histology GIT Block · MEDucation by Mattin Majid · Source: GIT Histology LG 3, Dr Hero K. Mustafa
Study alongside the lecture, not instead of it.